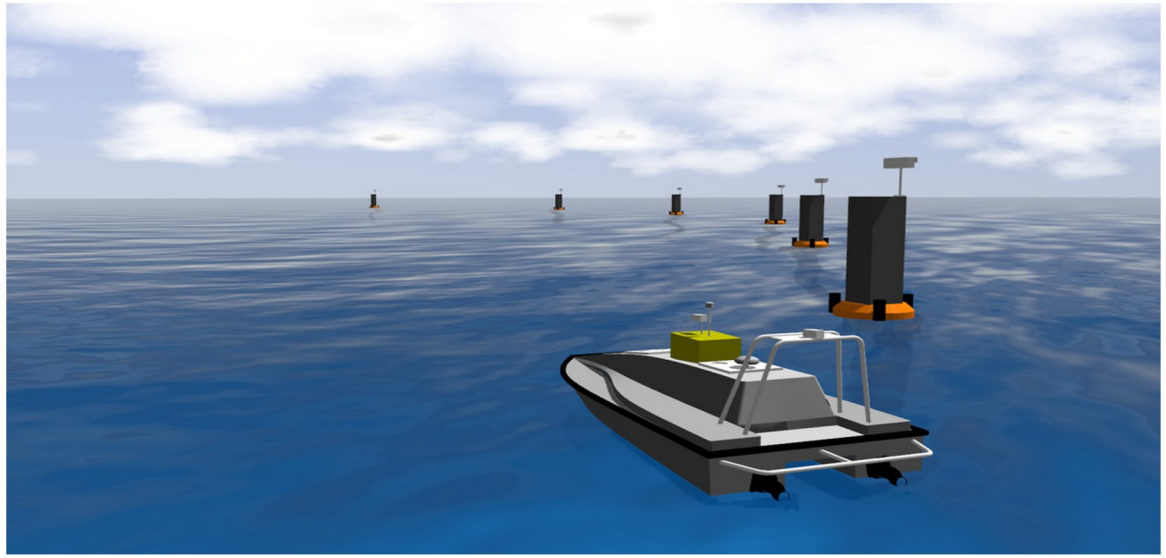
ISEP INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

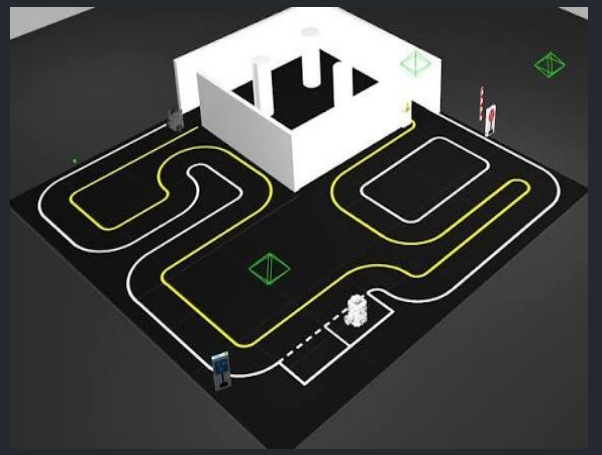
UTP Universidade Técnica do Porto

RSDIS | FINAL PROJECT PROPOSAL

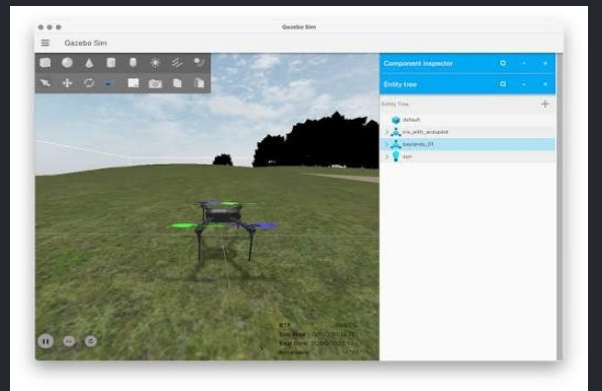
Multi-Robot Simulation Environment with ROS and GAZEBO

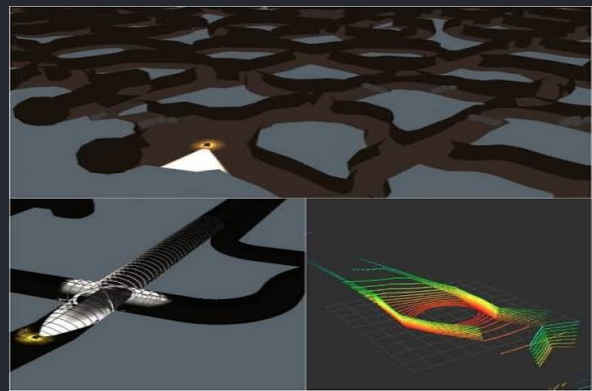
André Dias, Benedita Malheiro and Claude AI | Academic year 2026/27











Why Multi-robot?

KEY ADVANTAGE

Complementary sensing

Each robot is a mobile payload for different sensors. Together the team perceives more than any single robot, and fusing their data gives a richer picture of the scene.

UAV scout
RGB and thermal camera, GPS / IMU

UGV rescue
2D / 3D LiDAR, RGB-D camera, wheel odometry

Parallel coverage
Large areas are searched faster by splitting the work between robots.

Redundancy
If one robot fails, the others can continue the mission.

Specialisation
Aerial and ground robots take complementary roles.

Scalability
Several simple, low-cost robots can replace one complex machine.

Why 3D Simulators?

Safety

Test risky behaviours without endangering people or robots.

Lower cost

No hardware needed: try many robots and sensors for free.

Repeatability

Run identical experiments as often as needed and compare results.

Easy debugging

Access ground truth and log everything to find errors quickly.

Any scenario

Create hazardous or rare situations, such as disaster zones, on demand.

Faster development

Iterate quickly and run tests in parallel before using real robots.

With ROS and Gazebo, the same code tested in simulation can later run on the real robots.

Project objectives

Gazebo simulation scenario

Develop or reuse a 3D simulation scenario in Gazebo. The choice of the application scenario and of the challenge will be defined by the working group.

Multi-robot control and perception in ROS

Develop the ROS modules for multi-robot control and perception. Existing packages may be used, but the group must develop the modules that fulfil the objectives of the work.

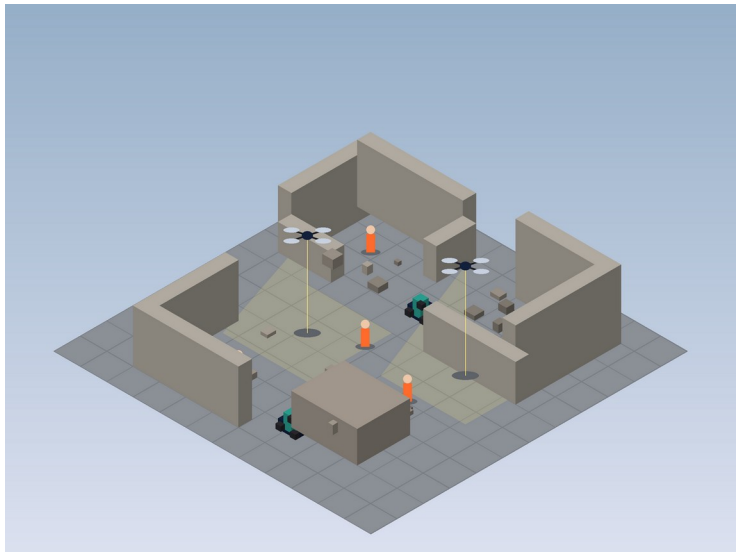
Interaction between robots

Ensure information sharing (maps, detections) and complementary tasks, for example UAVs scouting and UGVs confirming victims.

Report in IEEE paper format

Write a report in the form of a scientific paper, following the IEEE paper format, describing the work, the results and the conclusions.

Simulation environment preview



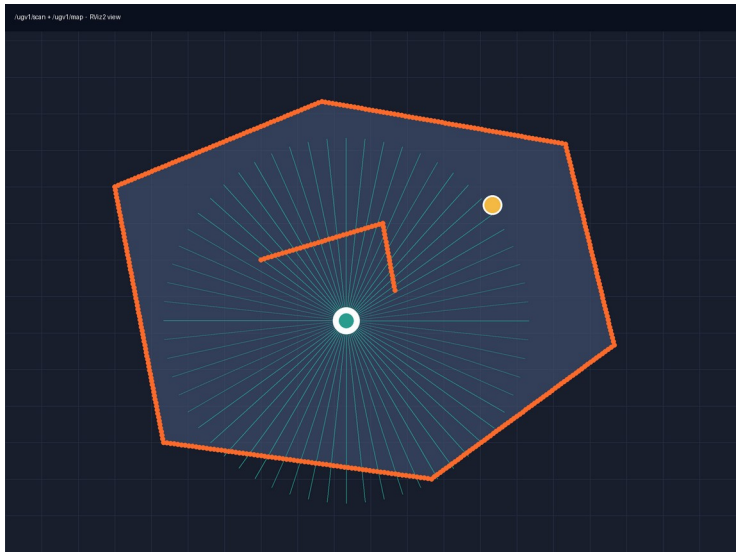
3D disaster-zone world

Gazebo world with walls, rubble, victims and the UAV + UGV team.



UAV camera view

Aerial images used by the detection node to find victims.

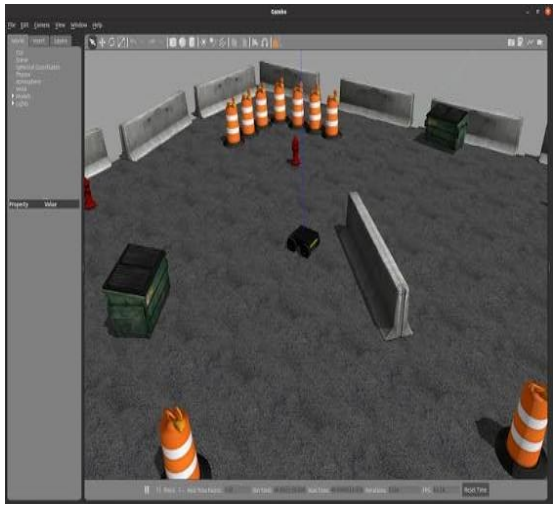


UGV LiDAR and map view

Scan and map shown in RViz2 for navigation and exploration.

Concept renders; to be replaced by screenshots of the final Gazebo and RViz2 simulation.

Example scenarios: ground robots



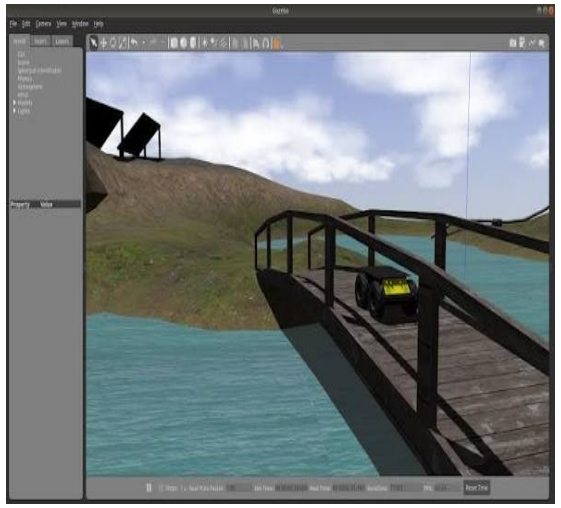
Urban obstacle course

Barriers and containers to test UGV navigation and avoidance.



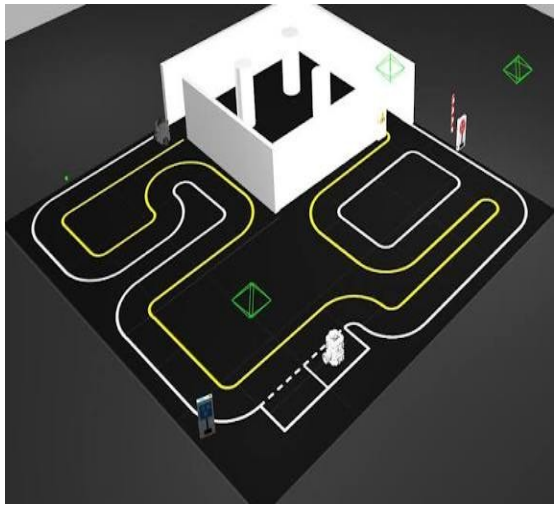
Outdoor rubble yard

Debris, sandbags and furniture: a cluttered area for ground search.



Rough terrain

Uneven ground and a bridge for a Husky-type UGV.

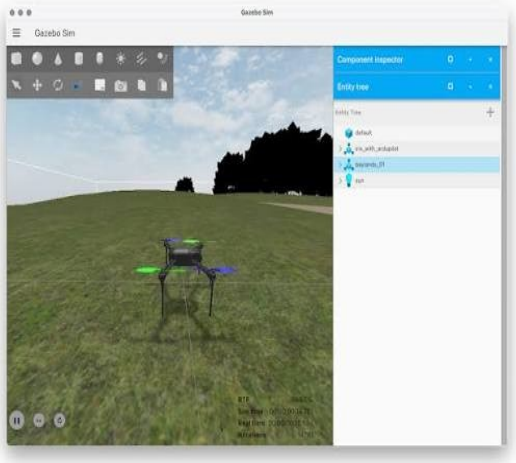


TurtleBot3 AutoRace

Track with lanes, signs and a tunnel; ready-made UGV navigation tests.

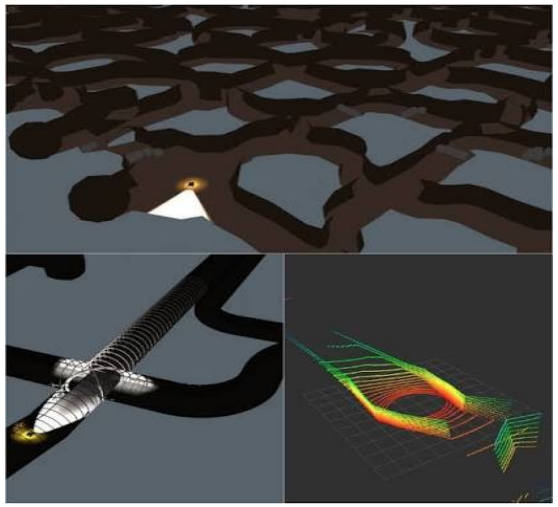
Illustrative examples of Gazebo environments (Gazebo community; TurtleBot3 AutoRace simulation by ROBOTIS).

Example scenarios: aerial, underground, maritime



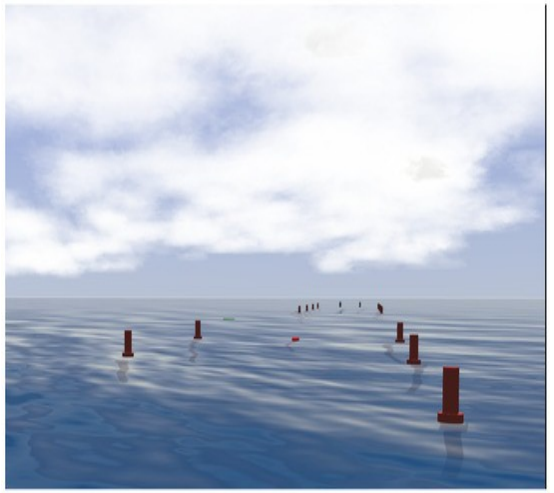
Aerial robot

Quadrotor in Gazebo Sim, a candidate UAV scout.



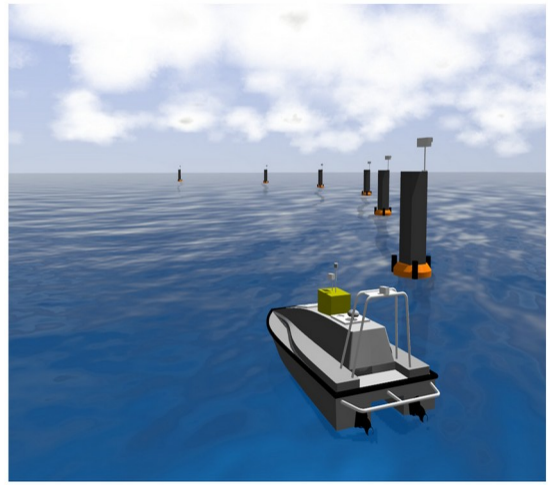
Tunnels and caves

Maze-like underground world with LiDAR data; useful for confined-space search.



Open-sea buoy field

Marine environment with obstacles; a basis for maritime search and rescue.



Surface vessel

Autonomous boat that could act as a rescue unit at sea.

Illustrative examples of Gazebo environments; maritime images from J. Mar. Sci. Eng. 2023, 11, 2058 (MDPI).

Heterogeneous robot team

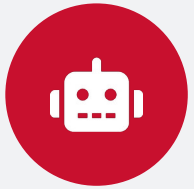


UAV scouts (x2)

Role Fast aerial coverage; spot victims from above and guide ground robots.

Sensors Downward RGB camera, IMU, GPS / odometry.

Control Quadrotor model (e.g. Hector or X3) with a waypoint controller.



UGV rescuers (x2)

Role Navigate cluttered terrain, reach victims and confirm detections.

Sensors 2D LiDAR, RGB-D camera, wheel odometry.

Control Differential-drive base (e.g. TurtleBot3 or Husky).

System architecture

Gazebo

3D world, physics and sensor plugins

ROS bridge

Gazebo and ROS topics and TF

ROS Workspace

Per-robot SLAM, Nav2, detection and custom nodes

Coordination

Map merge, task allocator, mission manager

Monitoring

RViz2, rosbag and metrics logger

NAMESPACED ROS INTERFACES

```
/uav1/camera/image_raw     /ugv1/scan     /ugv1/cmd_vel
/ugv1/map  ->  /merged_map     /victims/detections
```

Software components

Function	Approach	ROS tools
Mapping and localisation	SLAM on each robot; maps merged into one shared map	slam_toolbox, map merge
Navigation	Global and local planning with obstacle avoidance	Nav2
Exploration	Frontier-based search over assigned regions	Custom node, explore_lite
Victim detection	Marker, colour or object detection on camera images	OpenCV, apriltag_ros
Task allocation	Auction-based assignment of regions and victims	Custom coordinator node

Example Mission Workflow: Search and Rescue

EXAMPLE: SEARCH AND RESCUE

1

Deploy

Robots spawn at the base station; localisation and communications start.

2

Explore

UAVs and UGVs search their assigned regions with frontier-based exploration.

3

Detect

Camera nodes detect victims and publish estimated positions.

4

Allocate

The coordinator assigns the closest suitable robot to each victim.

5

Confirm

A UGV reaches the victim and its position is reported to the base.

Loop: while victims remain unconfirmed, robots re-plan and tasks are re-allocated.

NOW, YOUR PROBLEM

What is the mission workflow for your own scenario?

Evaluation plan

Simulation scenario

Application scenario developed or adapted in Gazebo.

Chosen challenge

Originality counts: out-of-the-box ideas will be valued.

ROS nodes developed

Nodes created by the group for control and perception.

Existing ROS packages

How existing ROS packages are used and integrated.

Solution developed

Overall multi-robot solution and how well it meets the objectives.

Quality of the IEEE paper

Clarity, structure and rigour of the report in IEEE paper format.

Deliverables and risks

Deliverables

- Gazebo world and robot models (SDF / URDF)
- ROS packages, launch files and README
- Demo video of the multi-robot mission
- Final report and presentation

Risks and mitigation

Heavy simulation load
Use simpler worlds and fewer sensors; run headless for experiments.

UAV control complexity
Start from a ready-made quadrotor plugin; fall back to a UGV-only team.

Integration delays
Test each module early and keep weekly integration milestones.

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Questions?

Expected outcome: a reproducible 3D Gazebo and ROS simulation in which a multi-robot team cooperates to accomplish a challenge chosen by the working group.

